

Sample End Semester Question Paper - 2024

Subject: Database Management Systems | Code: CS302 | Semester: 3
Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. What does the WHERE clause do in SQL?
2. Define a primary key.
3. State one requirement of 2NF.
4. Name the four properties of ACID.
5. What is a deadlock?
6. What is an index?
7. Define DBMS. State two advantages over file-based systems.
8. Name the three schema levels of the ANSI-SPARC architecture.
9. What is an E-R diagram? Name its three basic symbols.
10. State one difference between DDL and DML.

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain locking and concurrency control. Discuss serialisability.
2. Explain database security: authentication, authorisation and integrity.
3. Explain nested queries and correlated sub-queries with examples.
4. Explain the E-R model: entities, attributes, relationships and cardinality. Convert a sample E-R diagram into tables.
5. Explain SQL with DDL, DML, DCL and TCL, giving one example statement for each.
6. Write SQL queries using WHERE, GROUP BY, HAVING and JOIN for a given sample schema.
7. Explain normalisation: 1NF, 2NF, 3NF and BCNF with a worked example.
8. Explain transaction management and ACID properties with examples.
9. Explain deadlocks: conditions, prevention, avoidance and detection. Solve a Banker's algorithm example.
10. Explain views: types, creation and applications.